Fleet Game Demo Progress Report

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CSCE552 Game Development, Dr. JJ Shepherd

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I. General Progress Update

This week did not go as well as I had hoped. I had the game developer equivalent of writers block and felt overwhelmed at implementing all of the planned features despite not having other obligations this week. I wanted to make more progress than I have and development was very slow. I will probably have to update what features I implement for the final version given the rate of progress this past week. I also had to look up documentation or ask AI how to implement a feature more than I wanted to, but I have been taking it one feature at a time and trying to chunk the project. However, there is a playable training demo now with the basics of a shmup game loop and you can implement the loadout features I imagined, so it wasn't all bad. I am curious to know if my basic polygons are par for the course or if other students actually have good art.

II. Features Currently Implemented

1) Implemented movement that feels more like a starship flying.
2) Have buildings of random size spawn and come at the player.
3) Created turrets that shoot at player.
4) Created a background for the training scene. (I had to ask an AI for help with this.)
5) Added a targeting reticule and camera movement that made the game feel better. This still needs some work after playtesting.
6) Created a loadout screen that enforces the rules for choosing components.
7) Created art using PICO8 for the components and the ship status.

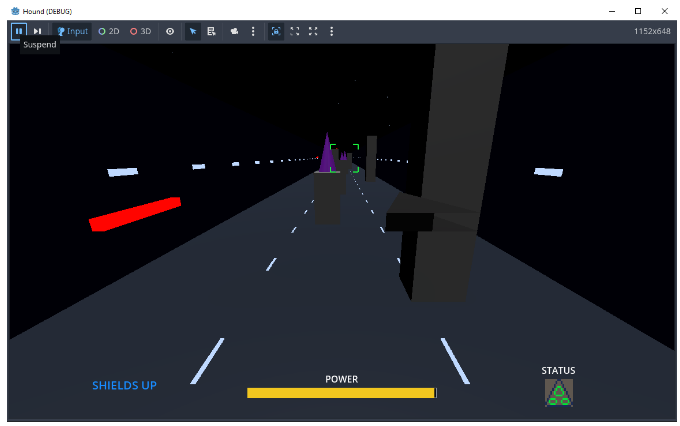

Fig. 1. screenshot of current game

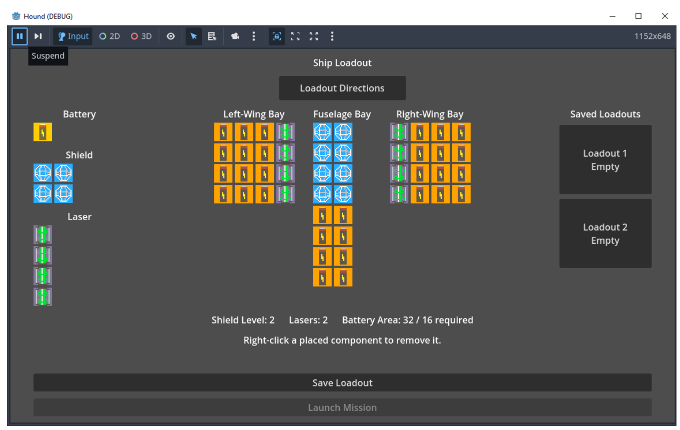

Fig. 2. loadout screen

8) The player can save up to two loadouts.
9) Recorded PEW and BOOM noises for firing lasers and lasers hitting objects. (It is literally just me making noises with my mouth.)
10) Created a warning noise when wing bays are damaged.
11) Drew art for the galaxy map, but have not implemented it.
12) The player can shoot enemy turrets and die from getting shot while shields are down or hitting a building. Right now the game just resets since only Training mode is working.
13) The scene tree for the game has grown, but the important scenes are now:
    a) Title
    b) Controls
    c) Loadout
    d) Training

III. Features To Be Implemented

Given my progress this week (or lack thereof), I am aiming for a minimally complete game loop of playing several levels.

1) Create enemy spaceships that move in and out of player view and attack.
2) Create asteroid objects that fly at player in space levels.
3) Create the galaxy map that allows the player to choose missions.
4) Create randomized levels instead of just infinite training.
5) Write the soundtrack for the game. (I plan to use PICO-8. I realized I can export .wave files from it.)
6) Hopefully, still have an open-ended structure where the player can attempt to play the “boss” level whenever they want, but playing other levels first to weaken the strength of the enemy forces.
7) Tracking number of ships and making a new pilot name whenever there is a death.
8) At least one on-rails planned level.

IV. Features Removed

1) I am removing bombs and missiles. The player will only have the laser system.
2) Fuel and component gathering. We will assume infinite fuel and components but still limit ships.
3) The retreat system for selecting a new ship with a different loadout. When the player dies they will either continue with a new ship or return to mission screen.
4) The roguelite elements that allow for upgrades.
5) Reducing the number of on-rails planned levels.

V. Problems Encountered

1) The knowledge and learning curve for Godot is very steep. I was overconfident last week after completing the tutorial and making a minimum 3D scene. As mentioned in the previous report, I still find the scene and node structure for creating game assets and code un-intuitive and I need to work to expand my knowledge of built-in tools so that I use what is available rather than trying to just code something. I had to ask an AI which node to use a lot.
2) I am doing my best with some PICO-8 2D pixel art for icons and basic polygons for all 3D art, but beyond this art ability I am not very happy with how the game looks. I am not a good enough artist to be a one-month solo dev.
3) General motivation. I crashed a bit this week after probably doing too much the last 3 weeks. Hopefully I do better this final week.

VI. Team Tasks

- **Taylor, July 24–27:** Create set-up for all randomly generated levels, create galaxy map and mission selection system.
- **Taylor, July 27–28:** Create at least one on-rails hand-crafted level to test the PathFollow node.
- **Taylor, July 29–31:** Finishing details, playtest and revise, submit game.
- **Clayton, July 20–31:** Create art and music as necessary to assist Taylor. Playtest and give feedback.

VII. Programming Tasks

- **Taylor, July 24–27:** Generating enemies and obstacles for randomly generated levels. Figure out how the path follow node interacts with player input.
- **Taylor, July 24–27:** Mission select and galaxy map logic.
- **Taylor, July 27–29:** Create as many levels as possible.
- **Taylor, July 29–31:** Finishing details, playtest and revise, submit game.

VIII. Art Tasks

At least this list is smaller than it was!

- **Taylor, July 24–27:**
  - Enemy Spaceship 1 Model
  - Enemy Spaceship 2 Model
  - Enemy Spaceship 3 Model
  - Laser
  - Asteroid Model
- **Taylor, July 27–29:**
  - Space Station Model
  - Planet Background
  - Improved Player Spaceship Model
  - Carrier Ship Model
  - Galaxy Map

IX. Testing Tasks

- **Taylor, July 24–29:** Conduct playtests focused on controls, objective clarity, interface readability, mission length, and difficulty.
- **Taylor, July 29–31:** Perform a final test and correct crashes, soft locks, and progression blockers.
- **Clayton, July 20–31:** Conduct playtests focused on controls, objective clarity, interface readability, mission length, and difficulty.